Multiscale waveform compatibility

1. Inputs

One event: H1 + L1

H1



L1



S **Short window**



−1.75 to +0.25 s

2048 Hz

L **Low frequency window**

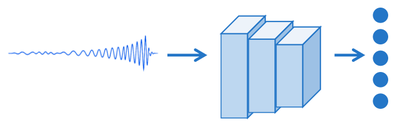


−15.75 to +0.25 s

20–80 Hz
256 Hz

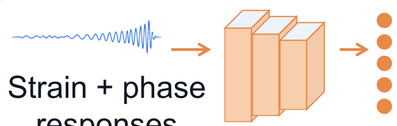
2. Event representations

S **Short waveform encoder**



e, M_c, q

S **Short waveform mass predictor**



Strain + phase responses

$p(M_c)$

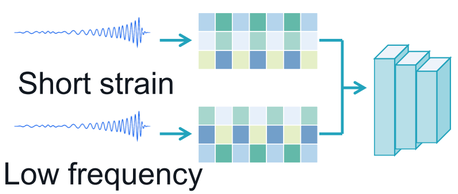
S **Template response mass predictor**



Mass-ordered template responses

$p(M_c)$

Multiscale joint predictor



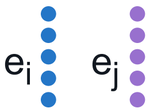
Short strain

Low frequency

$p(M_c, \eta, \chi_{eff})$

3. Pairwise comparison

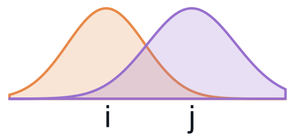
Representation compatibility



e_i e_j

Cosine similarity
Parameter differences

Mass compatibility from strain



i j

Distribution overlap

Mass compatibility from templates

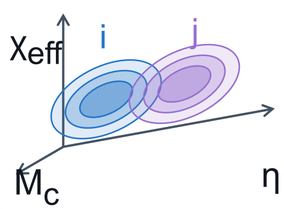


i
j

Training prior

Overlap + prior weighting

Joint parameter compatibility



χ_{eff} i j

M_c η

Joint overlap

4. Calibration

KDE calibration

Z_{short}

Finite-tail penalty

$\Delta_R = \gamma_R T_R$

Tail penalty + bounded increment

$\Delta_M = \gamma_M T_M + \beta_M I_M$

Tail penalty + bounded increment

$\Delta_J = \gamma_J T_J + \beta_J I_J$

$$Z_{wf} = Z_{short} + \Delta_R + \Delta_M + \Delta_J$$

Waveform score matrix